

# Moving Mass, Decreasing Energy

## Learning Probability Distributions with Wasserstein Gradient Flows

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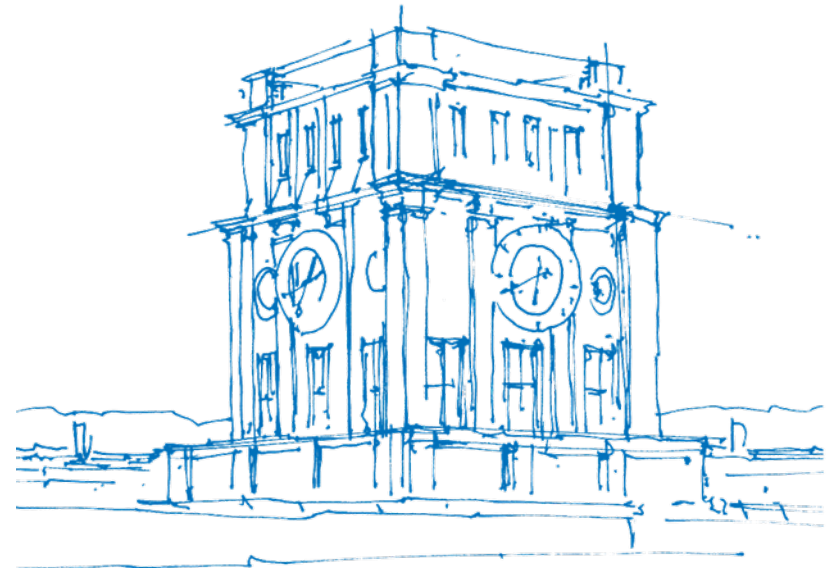
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# Motivation: Sampling as Optimization

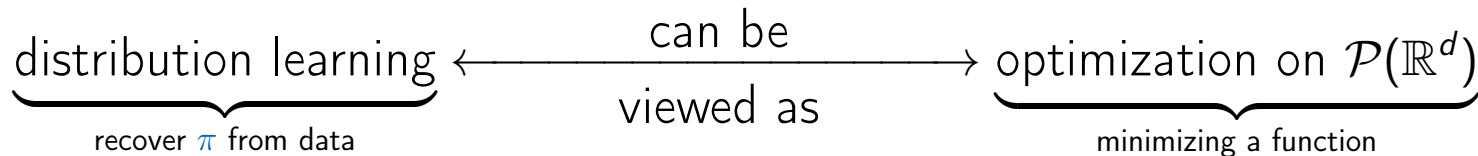
Goal: find points  $X_1, \dots, X_N \in \mathbb{R}^d$  approximating unknown probability distribution  $\pi \in \mathcal{P}(\mathbb{R}^d)$

Given:  $V: \mathbb{R}^d \rightarrow \mathbb{R}$  such that  $\pi$  has density  $\frac{1}{Z}e^{-V}$  OR samples  $Y_1, \dots, Y_M \sim \pi$ .

Solution:

- ① Design  $\mathcal{F}_\pi: \mathcal{P}(\mathbb{R}^d) \rightarrow [0, \infty) \cup \{\infty\}$  with  $\mathcal{F}_\pi(\mu) = 0$  iff  $\mu = \pi$ .
- ② Construct curve  $\mu: [0, \infty) \rightarrow \mathcal{P}(\mathbb{R}^d)$ ,  $t \mapsto \mu_t$  with  $\mathcal{F}_\pi(\mu_t) \rightarrow 0$ ,  $t \rightarrow \infty$ .
- ③ Approximate  $\mu_t$  by  $X_1(t), \dots, X_N(t)$ . Output  $X_1(T), \dots, X_N(T)$  for some large  $T > 0$ .

## TAKE-HOME 1 / 3:



# Refresher on Gradient Flows

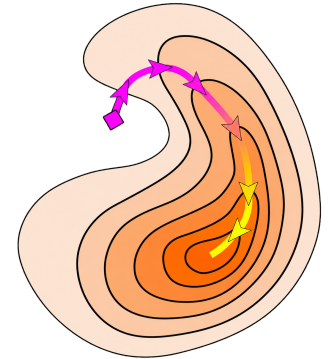
Goal: minimize „energy“  $F \in C^1(\mathbb{R}^d; \mathbb{R})$

Solution: Find curve  $x: [0, \infty) \rightarrow \mathbb{R}^d$ ,  $x \mapsto x_t$  satisfying

$$\frac{d}{dt}x_t =: \dot{x}_t = -\nabla F(x_t), \quad t > 0$$

Energy dissipation:

$$\frac{d}{dt}F(x_t) = \langle \nabla F(x_t), \dot{x}_t \rangle = \langle \nabla F(x_t), -\nabla F(x_t) \rangle = -\|\nabla F(x_t)\|^2 \leq 0.$$



## TAKE-HOME 2 / 3:

Gradient flow needs two ingredients:

- ① Energy  $F$       and      ② Geometry  $(\mathbb{R}^d, \langle \cdot, \cdot \rangle, \|\cdot\|)$ .

# How Do Probability Measures Move?

Take curve  $(\mu_t)_{t>0}$  such that  $\exists$  vector fields  $(v_t: \mathbb{R}^d \rightarrow \mathbb{R}^d)_{t>0}$  with  $\dot{X}_t = v_t(X_t)$  for any  $X_t \sim \mu_t$ . Then,

$$\partial_t \mu_t + \nabla \cdot (v_t \mu_t) = 0, \quad \text{weakly.}$$

**Derivation.** For any  $\phi \in C_c^\infty(\mathbb{R}^d)$ ,

$$\begin{aligned} \frac{d}{dt} \int_{\mathbb{R}^d} \phi(x) d\mu_t(x) &= \frac{d}{dt} \mathbb{E}_{X_t \sim \mu_t} [\phi(X_t)] = \mathbb{E}_{X_t \sim \mu_t} \left[ \frac{d}{dt} \phi(X_t) \right] \\ &= \mathbb{E}_{X_t \sim \mu_t} [\langle \nabla \phi(X_t), v_t(X_t) \rangle] = \int_{\mathbb{R}^d} \langle \nabla \phi(x), v_t(x) \rangle d\mu_t(x) \\ &= - \int_{\mathbb{R}^d} \phi(x) \nabla \cdot (v_t(x) \mu_t(x)) dx. \end{aligned}$$

This induces a geometry via the **Wasserstein metric**:

$$\frac{1}{2} W_2(\mu_0, \mu_1)^2 := \inf \left\{ \underbrace{\frac{1}{2} \int_0^1 \int_{\mathbb{R}^d} \|v_t(x)\|^2 d\mu_t(x) dt}_{\substack{\text{cost of moving } \mu_0 \text{ to } \mu_1 \\ \text{kinetic energy} = \frac{1}{2} \text{mass} \times \text{velocity}^2}} : \begin{array}{l} \partial_t \mu_t + \nabla \cdot (v_t \mu_t) = 0, \\ \mu_t|_{t=0} = \mu_0, \mu_t|_{t=1} = \mu_1 \end{array} \right\}.$$

**Thm.** Let  $\mathcal{P}_2(\mathbb{R}^d)$  be the space of probability measures with finite second moment. Then,

$(\mathcal{P}_2(\mathbb{R}^d), W_2)$  is a complete, separable metric space.

# Wasserstein gradient flows

**Goal:** Minimize  $\mathcal{F}_\pi: \mathcal{P}(\mathbb{R}^d) \rightarrow [0, \infty) \cup \{\infty\}$ .

Derivative in Euclidean space:  $DF(x)[v] = \langle \nabla F(x), v \rangle, \forall x, v \in \mathbb{R}^d$ .

**Defn.** Functional derivative of  $\mathcal{F}$  at  $\mu$  is  $\delta\mathcal{F}(\mu): \mathbb{R}^d \rightarrow \mathbb{R}$  and satisfies (up to add. const.)

$$\langle \delta\mathcal{F}(\mu), \phi \rangle_{L^2(\mu)} = \lim_{h \rightarrow 0} \frac{1}{h} (\mathcal{F}(\mu + h\phi) - \mathcal{F}(\mu)) \quad \forall \phi \in L^2(\mathbb{R}^d; \mu) \text{ admissible.}$$

The Wasserstein gradient flow of  $\mathcal{F}$  has the velocity field

$$v_t = -\nabla \delta\mathcal{F}(\mu_t).$$

**Why choose  $v_t$  like this?** Gradient flow means dissipating energy the fastest.

$$\begin{aligned} \frac{d}{dt} \mathcal{F}(\mu_t) &= \int_{\mathbb{R}^d} \delta\mathcal{F}(\mu_t) \partial_t \mu_t \, dx = - \int_{\mathbb{R}^d} \delta\mathcal{F}(\mu_t) \nabla \cdot (v_t \mu_t) \, dx \\ &= \int_{\mathbb{R}^d} \langle \nabla \delta\mathcal{F}(\mu_t), v_t \rangle \, d\mu_t(x) = \langle \nabla \delta\mathcal{F}(\mu_t), v_t \rangle_{L^2(\mathbb{R}^d; \mu_t)}. \end{aligned}$$

$\rightsquigarrow$  choose  $v_t = -\nabla \delta\mathcal{F}(\mu_t)$ : the velocity field is the steepest decrease of  $\mathcal{F}$ .

# Fokker–Planck as Wasserstein gradient flow

Choose  $\pi = \frac{1}{Z}e^{-V}$  and the **KL divergence**

$$\mathcal{F}_\pi(\mu) := \text{KL}(\mu \mid \pi) = \int_{\mathbb{R}^d} \log(\mu(x)) \, d\mu(x) + \int_{\mathbb{R}^d} V(x) \, d\mu(x) + \log(Z).$$

The velocity field is

$$\mathbf{v}_t = -\nabla \delta \mathcal{F}_\pi(\mu_t) = -\nabla (\log(\mu_t) + V + 1) = -\nabla \log(\mu_t) - \nabla V.$$

Hence, the WGF of  $\mathcal{F}_\pi$  is the Fokker–Planck equation

$$\partial_t \mu_t = -\nabla \cdot (\mu_t \mathbf{v}_t) = \nabla \cdot (\mu_t (\nabla \log(\mu_t) + \nabla V)) = \Delta \mu_t + \nabla \cdot (\mu_t \nabla V).$$

If  $\nabla V = 0 \rightsquigarrow$  heat equation Many other classical PDEs are Wasserstein gradient flows of suitable energies, e.g., Cahn–Hilliard, thin-film equations, liquid-crystal models, and chemotaxis equations.

# Particle Discretization

Sample  $X_1(0), \dots, X_N(0) \stackrel{\text{iid}}{\sim} \mu_0 \in \mathcal{P}_2(\mathbb{R}^d)$ . Assuming that  $\mu_t^N := \frac{1}{N} \sum_{j=1}^N \delta_{X_t^{(j)}}$ , solves the Wasserstein gradient flow

$$\partial_t \mu_t = \nabla \cdot (\mu_t \nabla \delta \mathcal{F}(\mu_t)), \quad (1)$$

we obtain *particle dynamics*

$$\dot{X}_t^{(j)} = \Gamma_t^{(j)}(X_t^{(1)}, \dots, X_t^{(N)}), \quad j \in \{1, \dots, N\} \quad \text{for some function } \Gamma.$$

## Problems:

- ①  $(\mu_t^N)_{t>0}$  might not solve (1).
- ② A typical velocity field such as  $\nabla \delta \mathcal{F}(\mu_t^N) = \nabla \log(\mu_t^N)$  might not be defined for empirical measures.
- ③ For  $\mathcal{F}_\pi = \text{KL}(\cdot \parallel \pi)$  with a continuous target  $\pi$ ,  $\mathcal{F}_\pi(\mu_t^N) = \infty \implies$  no informative finite loss landscape on empirical measures.

**Solution:** Regularization.

# Kernel regularization for Wasserstein gradient flows

**Choose:** kernel  $K: \mathbb{R}^d \times \mathbb{R}^d \rightarrow \mathbb{R}$  symmetric & positive definite, e.g.,  $K(x, y) = e^{-\frac{1}{2}\|x-y\|^2}$ .  
Uniquely determines the *reproducing kernel Hilbert space*  $H_K := \text{span}\{K(x, \cdot) : x \in \mathbb{R}^d\}$ .

## Regularize Energy

Instead minimize

$$\mathcal{F}_\pi^K(\mu) := \min_{\sigma \in M(\mathbb{R}^d)} \left\{ \mathcal{F}_\pi(\sigma) + \frac{1}{2} \left\| \int_{\mathbb{R}^d} K(\cdot, y) d(\mu - \sigma)(y) \right\|_{H_K}^2 \right\}.$$

**Advantage:**  $\mathcal{F}_\pi^K(\mu) < \infty$ , in particular for empirical measures  $\mu^N$ .

[Stein, Neumayer, Rux, Steidl, Anal. Appl., 2026]

## Regularize Geometry

Replace Wasserstein gradient velocity

$$v_t = -\nabla \delta \mathcal{F}(\mu_t)$$

by the non-local velocity

$$- \int_{\mathbb{R}^d} K(\cdot, y) \nabla \frac{\delta \mathcal{F}}{\delta \mu}(\mu_t)(y) d\mu_t(y).$$

**Advantage:** *interacting* particle flow; for  $\mathcal{F} = \text{KL}$ ,  $\nabla \log \mu$  isn't needed.

[Stein, Li, 2026a,b]



## Take-Home Messages

- ① Sampling can be viewed as optimization on the Wasserstein metric space.
- ② Gradient flow is determined by energy and geometry.
- ③ Kernel regularization can yield tractable particle approximations.

Thank you for your attention!

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